

CHIRPAN, Bulgaria

WMO: 156350

Lat: 42.200N

Lon: 25.333E

Elev: 584

StdP: 14.39

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.9	15.3	5.3	7.4	11.2	11.4	10.1	16.8	27.1	37.2	22.9	39.3	1.3	340	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.3	94.6	72.9	91.1	71.6	88.4	70.5	75.2	90.4	73.6	87.6	72.0	85.2	6.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.6	115.0	82.9	68.9	108.6	80.8	67.5	103.3	78.9	39.1	90.6	37.5	87.7	36.1	85.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	19.3	15.8	3.0	100.8	7.0	4.6	-2.0	104.2	-6.1	106.9	-10.1	109.5	-15.1	112.9
			2.6	78.8	7.0	3.0	-2.4	80.9	-6.5	82.7	-10.4	84.4	-15.5	86.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.3	32.8	37.1	45.5	53.7	62.5	69.9	74.5	74.3	66.4	55.6	44.0	33.8
	DBStd	16.34	8.45	8.92	7.40	6.49	5.69	5.48	4.62	4.96	5.64	7.13	7.18	8.79
	HDD50	1860	534	365	181	38	1	0	0	0	0	29	208	504
	HDD65	4811	999	782	606	340	115	18	1	2	50	299	632	967
	CDD50	3418	1	3	41	149	389	598	758	755	492	204	26	2
	CDD65	890	0	0	0	1	38	165	294	291	92	9	0	0
	CDH74	9976	0	0	4	41	502	1874	3424	3182	862	86	1	0
	CDH80	4023	0	0	0	4	103	698	1539	1428	240	11	0	0
Wind	WSAvg	4.9	3.9	5.0	6.4	6.6	5.6	4.8	4.9	4.7	5.2	4.1	4.4	3.7
Precipitation	PrecAvg	22.4	1.7	1.4	1.7	1.7	2.4	2.0	2.1	1.5	1.8	1.6	1.8	2.7
	PrecMax	29.9	5.3	3.0	5.5	3.8	5.7	5.1	6.5	3.4	5.0	3.3	5.4	6.6
	PrecMin	15.3	0.1	0.1	0.1	0.4	0.6	0.7	0.0	0.0	0.2	0.1	0.6	0.8
	PrecStd	3.8	1.2	1.0	1.3	0.9	1.3	1.2	1.7	1.0	1.4	1.0	1.2	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.1	64.8	72.9	79.2	86.6	94.8	101.0	98.9	90.5	82.1	68.2	58.5
		MCWB	48.4	51.8	56.9	62.8	69.4	71.7	75.7	72.8	70.3	67.7	57.8	52.4
	2%	DB	51.6	59.6	67.5	73.6	82.9	90.3	95.1	94.6	85.9	76.2	63.3	52.4
		MCWB	45.4	49.6	54.6	59.5	67.1	71.2	73.1	73.1	67.9	64.4	55.6	50.1
	5%	DB	47.2	54.9	63.6	70.1	79.6	87.1	91.2	91.0	82.6	71.9	59.0	49.0
		MCWB	43.5	47.8	52.8	58.0	65.3	70.2	71.7	71.7	66.3	61.4	53.6	46.9
	10%	DB	44.1	50.1	59.1	66.7	76.2	84.1	88.2	87.9	79.0	68.2	55.7	45.5
		MCWB	41.5	45.2	50.5	56.2	64.1	68.8	70.7	70.3	65.4	59.8	51.7	43.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.6	53.1	61.5	64.6	71.0	76.1	78.2	77.4	73.1	70.1	60.8	54.0
		MCDB	55.9	61.7	67.9	76.6	84.1	89.2	94.9	93.7	86.4	78.9	67.2	57.0
	2%	WB	46.7	50.4	55.7	60.9	68.5	73.5	75.2	75.1	70.5	66.1	56.8	50.4
		MCDB	50.1	57.9	64.8	71.3	80.4	86.7	91.1	90.3	82.3	73.8	61.2	52.2
	5%	WB	44.1	48.1	53.3	59.0	66.6	71.6	73.5	73.5	68.4	63.2	54.8	47.6
		MCDB	46.9	54.1	62.5	67.8	77.1	83.6	87.7	87.6	78.5	69.8	58.7	49.2
	10%	WB	41.6	45.6	50.9	57.1	64.7	70.0	71.9	71.7	66.5	60.9	51.9	43.6
		MCDB	44.1	50.0	58.8	65.1	74.1	81.7	85.3	84.9	75.8	66.6	55.4	45.4
Mean Daily Temperature Range	5% DB	MDBR	13.4	17.6	20.3	20.7	22.4	24.2	25.3	24.7	22.0	18.9	15.0	11.6
		MCDBR	20.7	27.3	28.4	27.4	28.8	29.4	30.4	29.7	27.8	25.3	21.3	16.0
		MCWBR	14.8	17.9	16.9	15.5	14.8	12.3	12.1	12.1	12.9	13.8	14.2	11.9
	5% WB	MCDBR	18.0	24.8	26.7	24.0	25.8	27.0	27.4	27.4	23.7	21.3	17.6	13.7
		MCWBR	13.3	17.2	16.8	14.5	14.5	12.4	12.6	12.8	12.0	12.6	12.3	10.6
Clear-Sky Solar Irradiance	taub	0.332	0.362	0.407	0.462	0.446	0.450	0.450	0.459	0.433	0.393	0.361	0.337	
	taud	2.364	2.285	2.189	2.038	2.165	2.191	2.199	2.170	2.215	2.309	2.357	2.359	
	Ebn at Noon	253	263	263	256	265	263	261	254	252	248	239	239	
	Edh at Noon	26	33	41	52	47	46	45	45	40	32	27	24	
All-Sky Solar Radiation	RadAvg	515	767	1098	1460	1802	1972	2058	1869	1383	912	584	445	
	RadStd	69	96	96	161	137	158	155	115	104	97	61	62	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (31 neighbors)		+0.83	N/A	N/A	N/A	+1.08	+1.08	-194	-281	+137	N/A

Nomenclature: See separate page